A Novel Approach to Design SAR-ADC: Design Partitioning Method

Prajit Nandi, Hirak Talukdar, Dhiraj Kumar, and Ashvin Kumar G. Katakwar

***Abstract*—This paper presents a successive approximation register analog-to-digital converter (SAR-ADC) design optimization platform. A novel ADC modeling approach, design partitioning method (DPM), is used in the proposed platform. It uses lookup tables along with conventional behavioral modeling technique, which captures the design parameters and process nonidealities. An individualistic as well as collective study of the impact of these parameters and nonidealities on the performance of SAR-ADC has been carried out. The proposed platform aims to achieve a perfect balance between architecture evaluation time (3.2% with respect to SPICE netlist) and accuracy in performance estimation (98% with respect to SPICE netlist). A comparative study between SPICE model, structural modeling, and the proposed modeling scheme (DPM) is reported here. As a case study, a 14-bit, 4.2MSPS SAR-ADC architecture is evaluated, designed, and verified.**

***Index Terms*—Architecture evaluation, design partitioning method (DPM), integrated circuit modeling, lookup table (LUT), successive approximation register analog-to-digital convertors (SAR-ADC), Verilog-A.**

I. Introduction

IN RECENT times, the time-to-market has been one of the most important motivating factor in an IC design cycle, which makes it very crucial to predict the overall performance and avoid possible cock-pit errors at the early stages of analog circuit design. A transistor-level implementation and analysis has been found highly time consuming [1] as well as limiting to perform yield analysis on large systems such as data converters. Hence, performing a design-for-manufacturability analysis becomes cumbersome. Behavioral models [2]–[10] of the desired mixed signal circuits have been proven to be the better solution to perform system-level analysis.

Recently, a successive approximation register analog-to-digital converter (SAR-ADC) design automation work [11] based on knowledge-based and simulation-based approach has been reported. In this paper, all the blocks are designed independently based on automated design algorithms for predefined block structures. It focuses on ∼10-bit resolution SAR-ADC design where power optimization is the main concern. Unlike bottom-up approach of the previous work, design partitioning method (DPM) mixes the top-down and bottom-up design approaches. Block-level simulation data can be examined independently through block-level test-benches as well as those data can be feed at top-level test-benches through lookup tables (LUTs) and their combined performance can be observed. In DPM, sampling switch and digital-to-analog converter (DAC) architecture are completely flexible unlike the previous design automation work. A MATLAB-based numerical tool to assist the design of charge redistribution DACs in SAR-ADCs has been presented recently in another reported work [12]. The overall SAR-ADC performance depends on combined nonidealities of capacitive DAC (CDAC) and sample–hold (S/H) network. Unlike DPM, the reported tool tries to optimize the CDAC independently. Though the effects of capacitive mismatch and parasitic capacitance are captured, voltage dependency of unit capacitors and switch resistances are not considered in the MATLAB model. As DPM works with SPICE device models, it is more compatible with analog design environment, which results in significantly lower design cycle time.

In the proposed modeling scheme along with conventional behavioral modeling, LUTs are generated for different modules to develop a faster and more accurate SAR-ADC design methodology. Details of LUT generation methodology are discussed in the following sections. It helps in optimal designing and analyzing the SAR-ADC with due consideration to the process nonidealities. Thus, it reduces design cycle time by avoiding unforced performance degradation after top-level integration. In the conventional structural modeling, all the blocks are behaviorally modeled [5]–[8] and organized as they are actually connected in the schematic. In the proposed modeling scheme, the whole SAR-ADC is partitioned into different blocks according to their functionality rather than their actual structure. Thus, it can be viewed as a technique to partition the whole design. Hence, the name DPM has been used. Root cause analysis can be done easily by making individual blocks real while keeping all the other blocks ideal. It aims at striking an optimized analysis across the design vectors, as illustrated in Fig. 1. The proposed platform uses Verilog-A [4] language to model different components of SAR-ADC.

This paper is organized as follows. Flowchart of the platform along with implementation details of different blocks and their integration are discussed in Section II. Section III covers the analysis and results. The advantages of the proposed model

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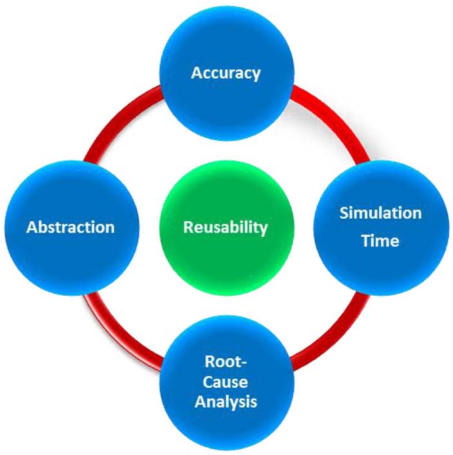


Fig. 1. Modeling optimization aspects.

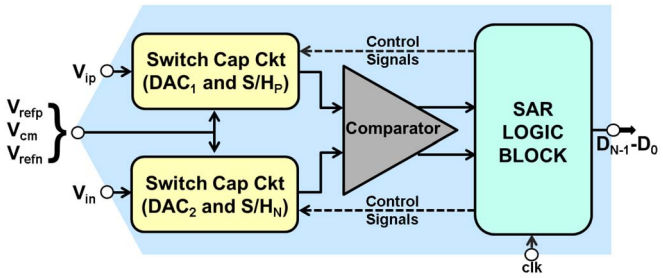


Fig. 2. Generic block diagram of SAR-ADC.

over other modeling platforms are discussed in Section IV. As a case study, a 14-bit, 4.2 MSPS SAR-ADC is discussed.

II. Implementation

Fig. 2 shows a generic block diagram of SAR-ADC which consists of switched capacitor network, comparator, and SAR-logic. The switched capacitor network functions as both S/H and DAC circuits [13]. The control signals control the DAC and S/H block. The aim of the proposed modeling scheme is to realize the SAR-ADC based on optimal mix of LUTs-based modeling and conventional behavioral modeling.

Operational flow of the proposed platform is shown in Fig. 3. Basic parameters V_{DD}, V_{SS}, unit resistance (R_{unit}), unit capacitance (C_{unit}), etc., are selected. Then the LUTs for S/H and DAC are generated by simulating SPICE netlists. The platform provides options to consider various nonideality parameters such as clock jitter, device noise, device mismatch (an equivalent to Monte Carlo analysis), etc., while generating data for the LUTs. Behavioral model of comparator and SAR logic are used (discussed in Sections II-C and II-D). Each functional sub-block of an SAR-ADC is modeled as a separate module having its own set of parameters. Post simulation extraction of ADC parameters is also a part of the platform. Details of modeling techniques of the major sub-blocks are discussed here.

A. Sample and Hold

As shown in Fig. 4(a), S/H model takes different nonidealities in consideration. In general, CDAC itself is used as S/H

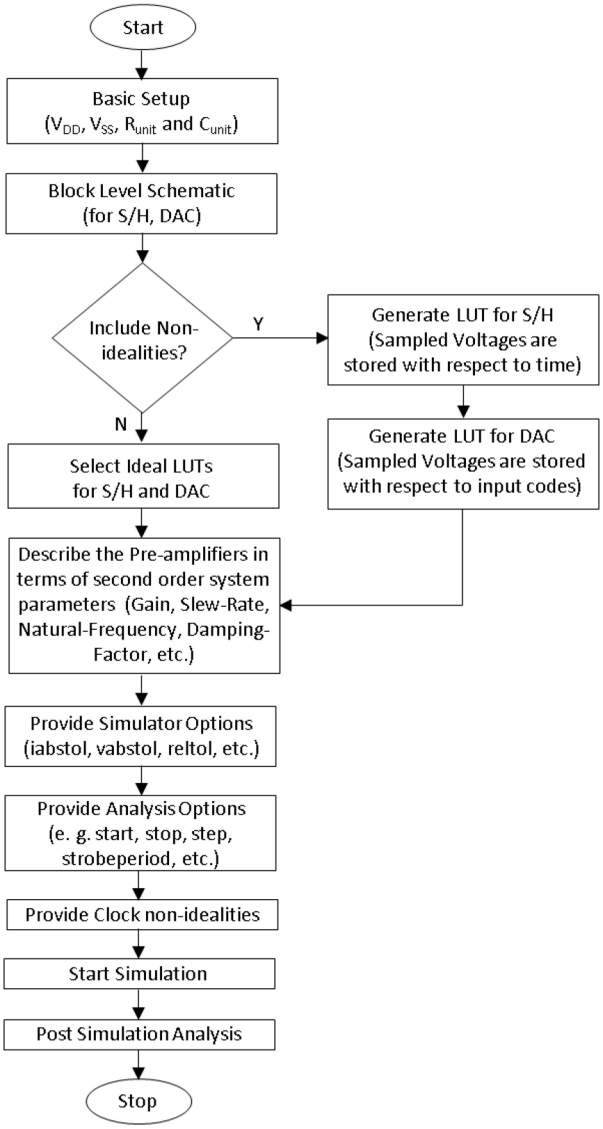


Fig. 3. Flowchart of the proposed SAR-ADC design platform.

circuit in SAR-ADC. Bottom-plate sampling technique [13] is employed in CDAC-based S/H circuits. For simplicity, single-ended bottom-plate sampling mechanism is shown in Fig. 4(b). The clock (φ) and delayed clock (φ_d) are used for bottom-plate sampling. Output voltage of S/H circuit can be described by (1) and (2) in ideal scenario. Based on rms jitter of the sampling clock, half-cycle jitter [14] is modeled. Spread of sampling clock transition edges is shown in Fig. 4(c) (for 5000 cycles).

$$V_{out} = V_{cm} \text{ (Sampling phase)} \quad (1)$$

$$V_{out} = 2 * V_{cm} - V_{in} \text{ (Hold phase)} \quad (2)$$

$$R_{on_sampling} = f_1(R_0, V_{in}, V_{DD}, T) \quad (3)$$

where V_{out} is the output voltage of S/H, V_{in} is the input voltage, V_{cm} is the common mode reference voltage, $R_{on_sampling}$ is the on-resistance of the input sampling switch, R_0 is on-resistance of sampling switch at zero-input, nominal temperature (27 °C) and nominal supply (2.5 V), and T is the absolute temperature.

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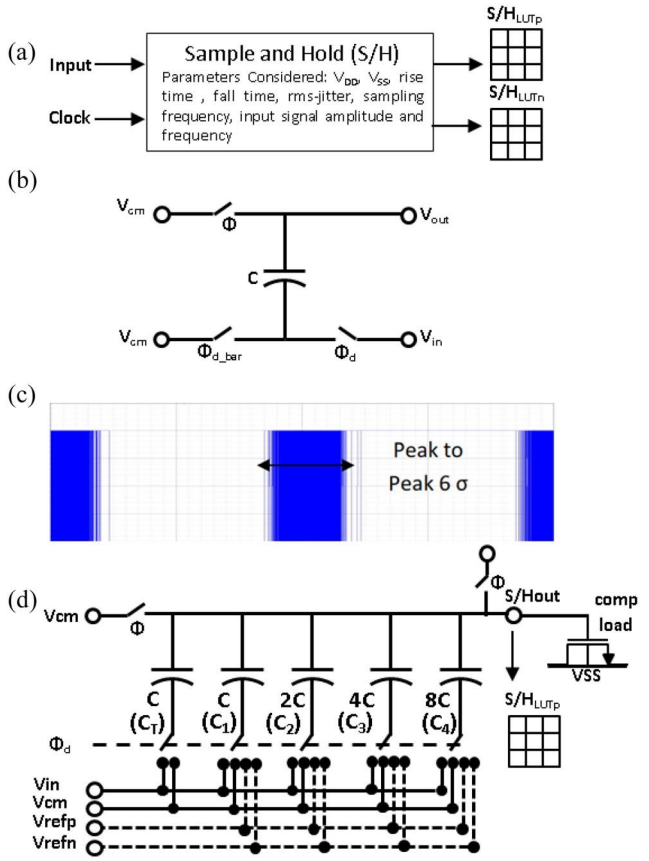


Fig. 4. (a) S/H port diagram. (b) Bottom-plate sampling mechanism (single-ended). (c) Spread of sampling clock transition edges (σ stands for rms value of the uncorrelated clock-jitter). (d) Structure of S/H corresponding to a 4-bit CDAC.

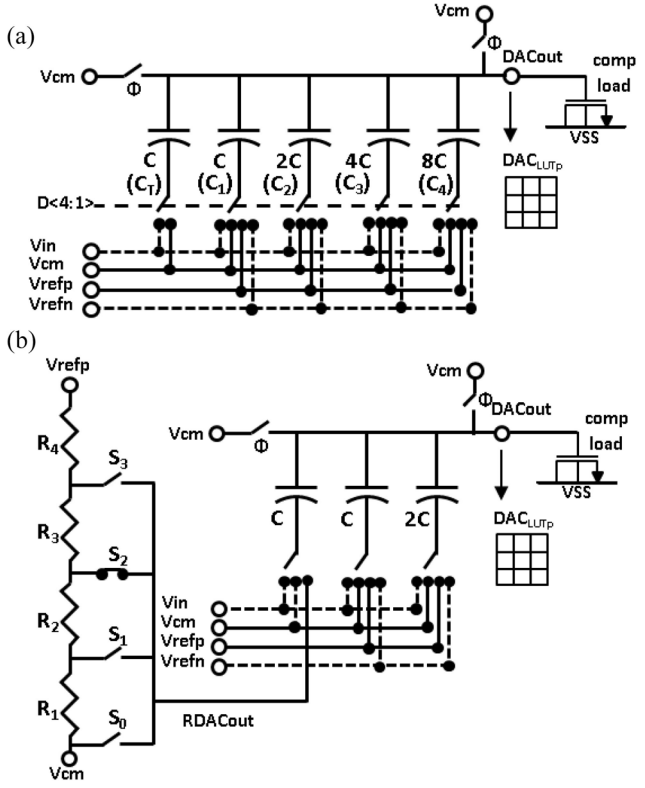


Fig. 5. (a) Structure of a 4-bit CDAC. (b) Structure of a 4-bit hybrid DAC (CDAC for MSB two bits and resistive DAC for LSB two bits).

For LUT generation, input-signal amplitude (A_{in}) and frequency (F_{in}) are selected. F_{in} is selected for coherent sampling [15], [16]. Rise time, fall time, and rms jitter are added to the sampling clock. Voltage and temperature dependency sampling switch on-resistance [described by (3)] and parasitic at different nodes of sampling network are considered. Top plates of the sampling capacitor arrays are loaded with the input differential pair of MOS devices of the comparator, as shown in Fig. 4(d). In the bottom-plate sampling mechanism, sampled voltages across capacitor arrays are stored into LUTs (S/H_{LUTp} and S/H_{LUTn}). One set of data get stored in every conversion cycle for differential S/H blocks. Settling behavior of S/H block is captured. The stored data in the LUTs are more than $(N+3)$-bit accurate (where N is the resolution of ADC) as they are generated by simulating SPICE models. It ensures less than 0.125-LSB top-level simulation error. The simulations become fast as standalone S/H block is simulated. For S/H circuit, the stop time of the block-level transient simulation is equal to the stop time of the top-level simulation.

B. Digital to Analog Converter

Conventionally, CDAC is used as DAC in SAR-ADC as the same DAC can be used as S/H circuit. As shown in Fig. 5(a), CDAC consists of an array of capacitors. During the conversion time, bottom plates of those capacitors can be connected to the common-mode reference voltage (V_{cm}) or positive reference voltage (V_{refp}) or negative reference voltage (V_{refn}) depending on the intermediate bit-pattern generated by SAR-logic. Bottom plate of the terminating capacitor (C_T) [17], [18] is connected to only V_{cm} (though half portion of the terminating capacitor can be connected to V_{refp} or V_{refn} to make the mean of the quantization error zero at the output of ADC). If required, scaling capacitor (C_S), hybrid DAC, charge redistribution DAC, DAC calibration, redundancy, and nonbinary algorithm [19]–[23] can also be incorporated in the proposed modeling platform by modifying the configuration of the DAC

$$V_{\text{out_}p_C} = V_{\text{cm}} + \sum_{i=1}^{N} \frac{C_i \times D_i}{C_{\text{tot}}} \times \left(V_{\text{refp}} - V_{\text{cm}}\right) + e_{\text{Cp}} \tag{4}$$

$$V_{\text{out_}n_C} = V_{\text{cm}} + \sum_{i=1}^{N} \frac{C_i \times D_i}{C_{\text{tot}}} \times \left(V_{\text{refn}} - V_{\text{cm}}\right) + e_{\text{Cn}} \tag{5}$$

where

$$e_{\text{Cp}}, e_{\text{Cn}} = f_2(\sigma_C, \tau_C, \text{clk}_{\text{FT}}, \text{clk}_{\text{CI}}, Q_{\text{LEAK}}, \text{COMP}_{\text{LOAD}}, N_{\text{KT/}C}) \tag{6}$$

$$\text{clk}_{\text{FT}} = V_{\text{CLK}} * \frac{C_{\text{gs_sw}}}{C_{\text{tot}}} \tag{7}$$

$$\text{clk}_{\text{CI}} = \frac{W_{\text{sw}} L_{\text{sw}} C_{\text{OX}}}{2} * (V_{\text{CLK}} - V_{\text{REF}} - V_{\text{TH}}) * \frac{1}{C_{\text{tot}}} \tag{8}$$

$$N_{\text{KT/C}} = \frac{kT}{C_{\text{tot}}} \tag{9}$$

$$V_{\text{out_}p_R} = V_{\text{cm}} + \left(\sum_{i=Q+1}^{N} \frac{C_i \times D_i}{C_{\text{tot}}} + \sum_{j=0}^{K} \frac{R_j}{R_{\text{tot}}} \right) \times \left(V_{\text{refp}} - V_{\text{cm}}\right) + e_{\text{Rp}} \tag{10}$$

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$$V_{\text{out_}n_R} = V_{\text{cm}} + \left(\sum_{i=Q+1}^{N} \frac{C_i \times D_i}{C_{\text{tot}}} + \sum_{j=0}^{K} \frac{R_j}{R_{\text{tot}}} \right) \times (V_{\text{refn}} - V_{\text{cm}}) + e_{\text{Rn}} \quad (11)$$

where

$$K = \sum_{x=1}^{Q} 2^{(x-1)} \times D_x \quad (12)$$

$$e_{\text{Rp}}, e_{\text{Rn}} = f_3(\sigma_R, \tau_R, \text{clk}_{\text{FT}}, \text{clk}_{\text{CI}}, \text{Ron}_{\text{sw}}, N_{0T}) \quad (13)$$

$$\text{Ron}_{\text{sw}} = f_4(R_0, V_{\text{REF}}, V_{\text{DD}}, T) \quad (14)$$

$$N_{0T} = 2^{Q*} 4kTR_{\text{unit}}. \quad (15)$$

Output voltages of an N-bit generic CDAC can be described by (4) and (5), where, $V_{\text{out_}p_C}$ and $V_{\text{out_}n_C}$ are the differential output voltages of CDACs, V_{refp} and V_{refn} are the positive and negative reference voltages, respectively, C_i is the capacitance value of the capacitor at ith position, D_i is the value of ith bit (1 or 0) generated by SAR logic, C_{tot} is the total capacitance of the capacitor array, N is resolution of CDAC (four bits), and e_{Cp} and e_{Cn} are the errors in CDAC outputs.

Different components of e_{Cp} and e_{Cn} are shown in (6). σ_C represents the standard deviation (SD) of the capacitor mismatch (device mismatch), τ_C is the time constant of the switch capacitor network (settling error), clk_{FT} (7) and clk_{CI} (8) are the effects of clock feedthrough and charge injection, respectively (output glitches due to switches), Q_{LEAK} represents the charge leakage from capacitors, $\text{COMP}_{\text{LOAD}}$ is the comparator load, and $N_{kT/C}$ is the kT/C noise (9) [24], [25]. k is the Boltzmann's constant (1.38×10^{-23} J/K).

In the hybrid DAC as shown in Fig. 5(b), CDAC is used for MSB bits and RDAC is used for LSB bits. Its output voltages can be described by (10) and (11), where, $V_{\text{out_}p_R}$ and $V_{\text{out_}n_R}$ are the differential output voltages, R_j is the resistance value of the resistor at jth position, R_{tot} is the total resistance of resistor ladder, K is the index of the on-switch (12), Q is the resolution of resistive subDAC (two bits), and e_{Rp} and e_{Rn} are the errors in the RDAC outputs.

Different components of e_{Rp} and e_{Rn} are shown in (13). σ_R represents the SD of the resistor mismatch (device mismatch), τ_R is the time constant of the switching network (settling error), Ron_{sw} is the nonlinear on-resistance of output switch (14), N_{0T} is the power spectral density of the thermal noise generated by resistor string (15) [26], [27].

Design parameters can be mathematically modeled by using functions f_1–f_4, but in reality those parameters are highly nonlinear and technology-node dependent. In the proposed platform, block-level SPICE models are simulated instead of using the functions to extract the accurate numerical points. Similar to the S/H, in the case of DACs, top plates of the capacitor arrays are loaded with the input differential pair of MOS devices of the comparator, as shown in Fig. 5. DAC output voltages are stored in LUTs (DAC_{LUTp} and DAC_{LUTn}) with respect to input codes.

In the case of the DACs of SAR-ADC, any input bit pattern is presided by a typical bit pattern only, as shown in Fig. 6(a) for a 3-bit SAR-ADC. The searching algorithm is analogous to a binary tree. Every node of a binary tree (input bit pattern) has only one parent node (previous bit pattern). To capture the memory effect (impact of the previous phase on

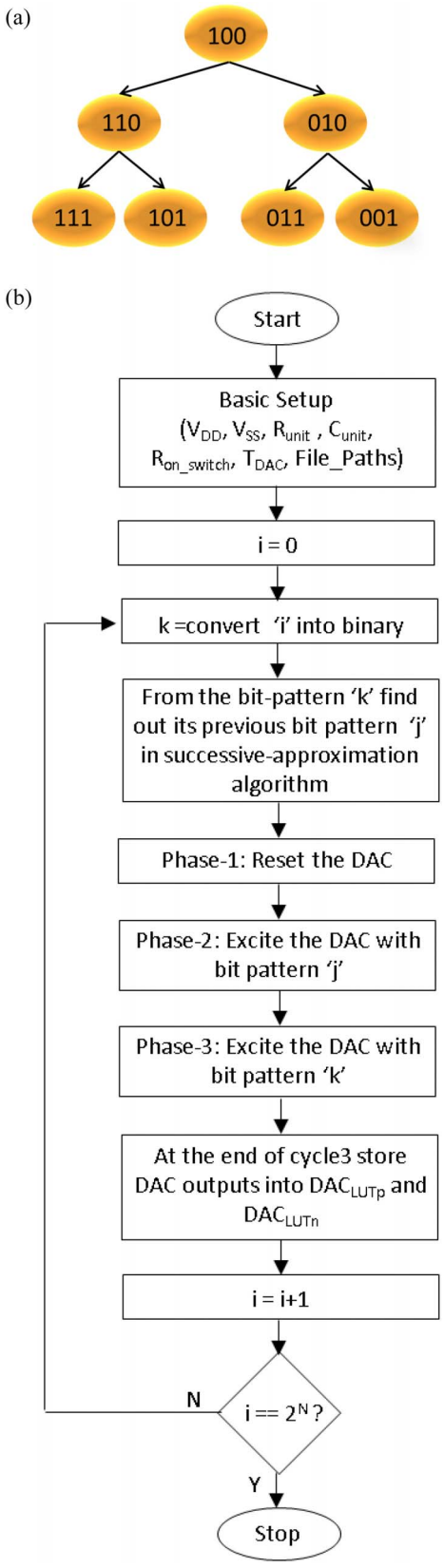


Fig. 6. (a) Binary tree for 3-bit SAR-ADC. (b) Flowchart for DAC LUT generation methodology.

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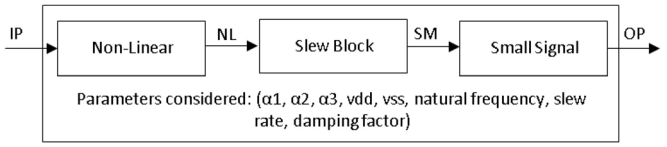


Fig. 7. Block diagram of preamplifier model.

the very next phase) of DACs, LUTs are generated by following the flowchart described in Fig. 6(b). In phase 1, DACs are reset. Followed by phase 2, where DACs are excited with the previous bit pattern to capture the memory effect. A separate algorithm based on the binary tree is used to search the previous bit pattern corresponding to the current bit pattern. In phase 3, DACs are excited with current bit pattern. At the end of phase 3, DAC output voltages corresponding to current bit-pattern are stored in LUTs. Simulations for DACs are done by considering input code as a parameter.

The DACs can be made up with resistors, capacitors, and switches modeled in Verilog-A or devices from technology library. For DAC simulations, SDs of capacitor and resistor mismatches; parasitic at different nodes of capacitors, resistors, and switches; on–off resistance of switches; voltage dependency of on-resistance of switches, transient glitches, capacitor leakage, etc., nonidealities can be considered optionally. As SPICE models are used for DAC simulations with memory effect, here also results are more than $(N + 3)$-bit accurate (where N is the resolution of ADC).

C. Comparator

In order to ensure 0.5-LSB comparator resolution error, preamplifiers with output offset cancellation followed by latch architecture [28], [29] are used in 14-bit SAR-ADC. Different stages of comparator are behaviorally modeled. The comparator output can be generated by solving few multivariable time-independent equations only. It makes the comparator model faster by reducing the number of iterations. Modeling methodologies of the different sections of comparator are discussed here.

1) Preamplifier: As shown in Fig. 7, the preamplifier is modeled as a system comprising of three sub-blocks, i.e., nonlinearity block, slew block, and small-signal block.

Three blocks collectively produce the behavioral equivalence of a real preamplifier. The model captures the impact of parameters such as slew-rate limitation, bandwidth limitation, dc offset, gain limitations, and device mismatch. Preamplifiers nonlinearity is realized through the nonlinear block using

$$A_{\text{nonlinearOut}} = \alpha_1{}^*(\text{input}) + \alpha_2{}^*\left[(\text{input})^2\right] + \alpha_3{}^*\left[(\text{input})^3\right] \tag{16}$$

where "$A_{\text{nonlinearOut}}$" (NL) is the dc output of the nonlinear block for the applied input signal (IP). The signal (NL) is then passed through slew block, which imitates the amplifier slew-rate limitations as per the flowchart as shown in Fig. 8. Here "SRO" is the voltage reached by slew block, "TimeRequired" is the slewing time of the amplifier, "HoldTime" is the time available for that preamplifier, and "timeLeft" (T_L) is the available time for small-signal block. The output of slew

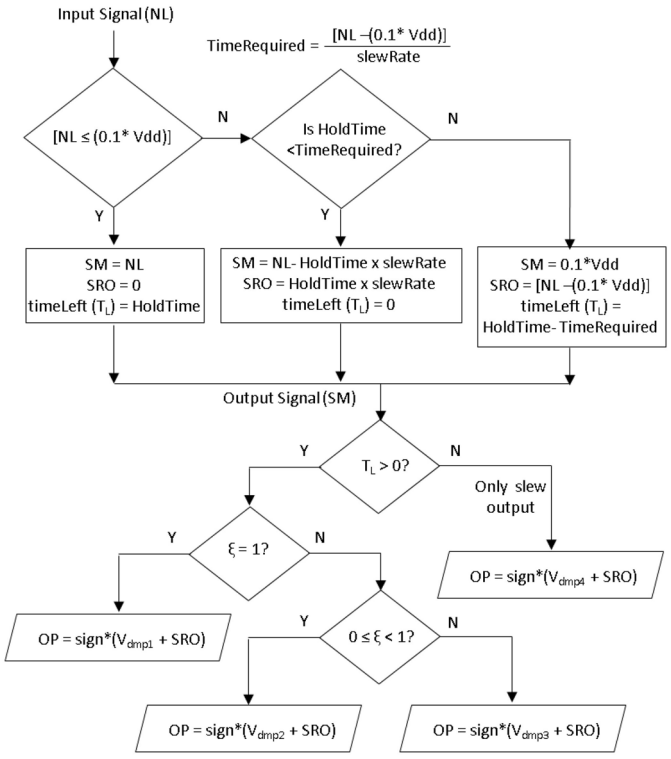


Fig. 8. Flowchart of the modeled slew block and small-signal block.

block (SM) is passed to the small-signal block, which realizes amplifier settling and bandwidth limitations using the characteristics of small-signal unit-step time response to a second-order system. Parameters of the small-signal block are damping ratio "ξ" and natural frequency of oscillation "ω_n." Output of the small-signal block is evaluated as

$$V_{\text{dmp1}} = \text{SM} \times \left[1 - \left\{e^{(-\omega_n \times T_L)}\right\} \times \{1 + (\omega_n \times T_L)\}\right] \tag{17}$$

$$V_{\text{dmp2}} = \text{SM} \times \left[1 - \left\{e^{\frac{(-\xi \times \omega_n \times t)}{\sqrt{(1-\xi^2)}}}\right\} \times \left\{\sin\left(\omega_n \times \sqrt{(1-\xi^2)} \times t + \arctan\frac{\sqrt{1-\xi^2}}{\xi}\right)\right\}\right] \tag{18}$$

$$V_{\text{dmp3}} = \text{SM} \times \left[1 - \left\{\frac{\left(-\xi + \sqrt{\xi^2-1}\right)}{2 \times \sqrt{\xi^2-1}} \times e^{\left(\left(-\xi - \sqrt{\xi^2-1}\right) \times \omega_n \times t\right)}\right\} + \left\{\frac{\left(-\xi - \sqrt{\xi^2-1}\right)}{2 \times \sqrt{\xi^2-1}} \times e^{\left(\left(-\xi + \sqrt{\xi^2-1}\right) \times \omega_n \times t\right)}\right\}\right] \tag{19}$$

$$V_{\text{dmp4}} = 0. \tag{20}$$

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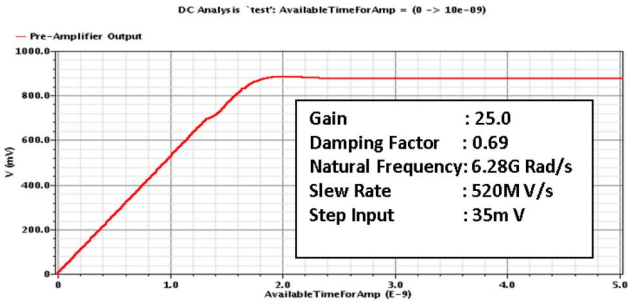


Fig. 9. Transient response of the modeled preamplifier.

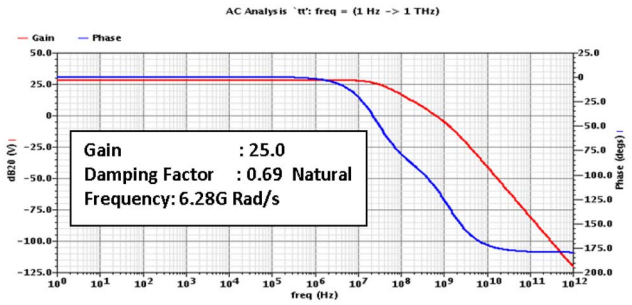


Fig. 10. Frequency response of the modeled preamplifier.

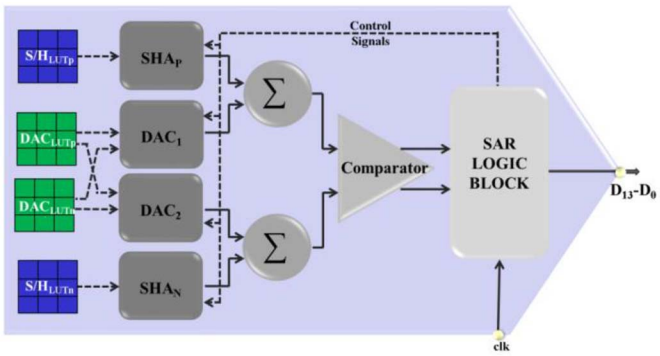


Fig. 11. Proposed modeling scheme of SAR-ADC top level: DPM.

The transient analysis of the preamplifier model can be done by varying the time available for that preamplifier (HoldTime). AC analysis of the model can also be done to capture the small-signal behavior of the model. The final output (OP) has similar characteristics of a real amplifier, examples are shown in Figs. 9 and 10. The platform also presents scope to have standalone preamplifier modeling and testing. It helps in deriving preamplifier specifications.

2) Latch: Calculated preamplifier output voltage (OP) is applied to the latch model. Parameters for the latch model are T_{Latch} (time available for the latch to take a decision) and τ_L (time constant of latch). Output of comparator V_{comp} is calculated based on the conventional latch characteristic equation, as shown in (21) [30]–[33]. Optionally, the latch offset can be included in the model

$$V_{\text{comp}} = \text{OP} \times e^{\left(\frac{T_{\text{Latch}}}{\tau_L}\right)}. \tag{21}$$

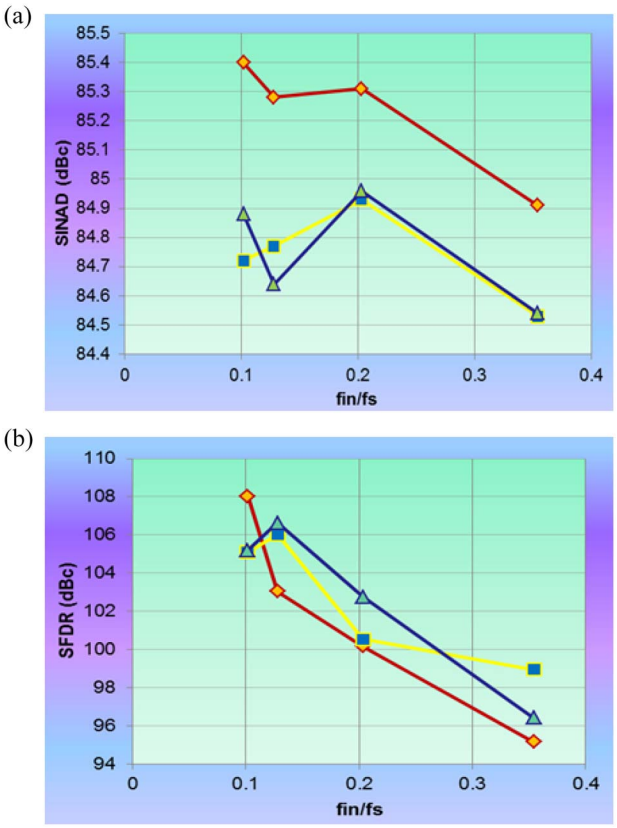


Fig. 12. (a) SINAD versus F_{in} plots. (b) SFDR versus F_{in} plots (red: SPICE netlist, yellow: structural model, and blue: DPM).

D. SAR Logic Block

SAR logic block is modeled in Verilog-A [34] to save simulation time. SAR logic generates all the control signals for switches and clocks for offset cancellation, latch and D flip-flops. The whole conversion-time is subdivided into 17 cycles. First three cycles are used for sampling and remaining 14 cycles are used for 14-bit conversion. For the 4.2 MSPS ADC, a 71.4 MHz (17×4.2 MSPS) clock is used. At the end of conversion (EOC), SAR logic provides an N-bit ADC output along with an EOC signal.

E. SAR-ADC Top-Level: Design Partitioning Method

In the top level, S/H circuits and DACs are implemented as Verilog-A modules. S/H modules produce outputs from pregenerated LUTs. Outputs of S/H blocks are function of time. DAC modules produce outputs corresponding to the input bit pattern generated by SAR logic. DAC modules also use the data from LUTs which are already prepared by simulating SPICE netlist of standalone DACs. Outputs of S/H modules and DAC modules are summed and the summer outputs go to the comparator model. As per the comparator outputs, SAR logic takes bit by bit decision and produces appropriate control signals.

As shown in Fig. 11, in the proposed modeling scheme each block is realized separately (i.e., separate S/H and DAC, though in reality they are a single block with dual functionality). It results in easy root cause analysis.

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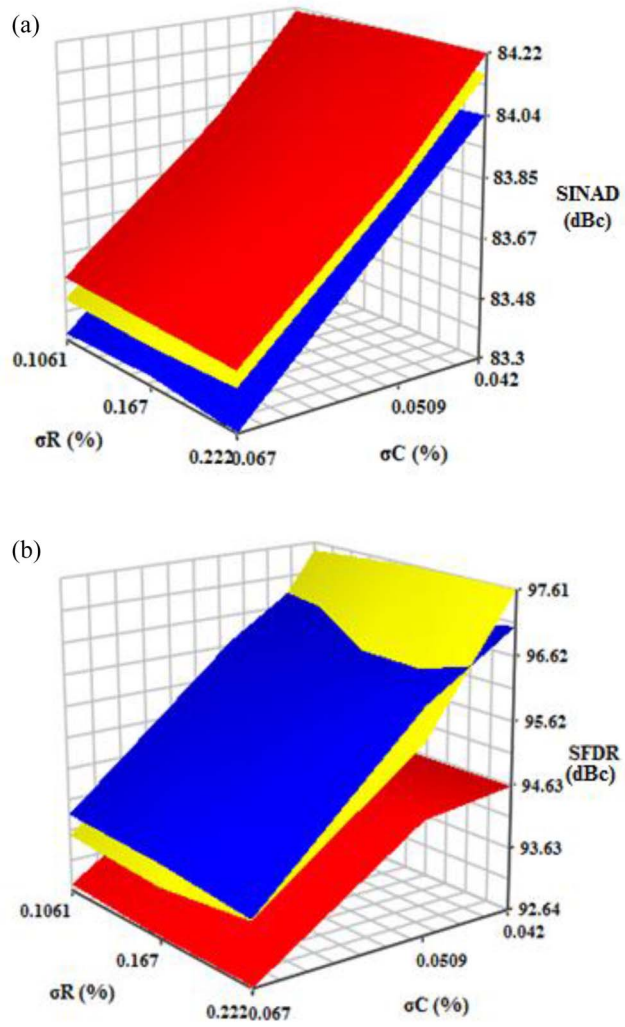


Fig. 13. (a) SINAD versus (σ_R–σ_C) plots. (b) SFDR versus (σ_R–σ_C) plots (red: SPICE netlist, yellow: structural model, and blue: DPM).

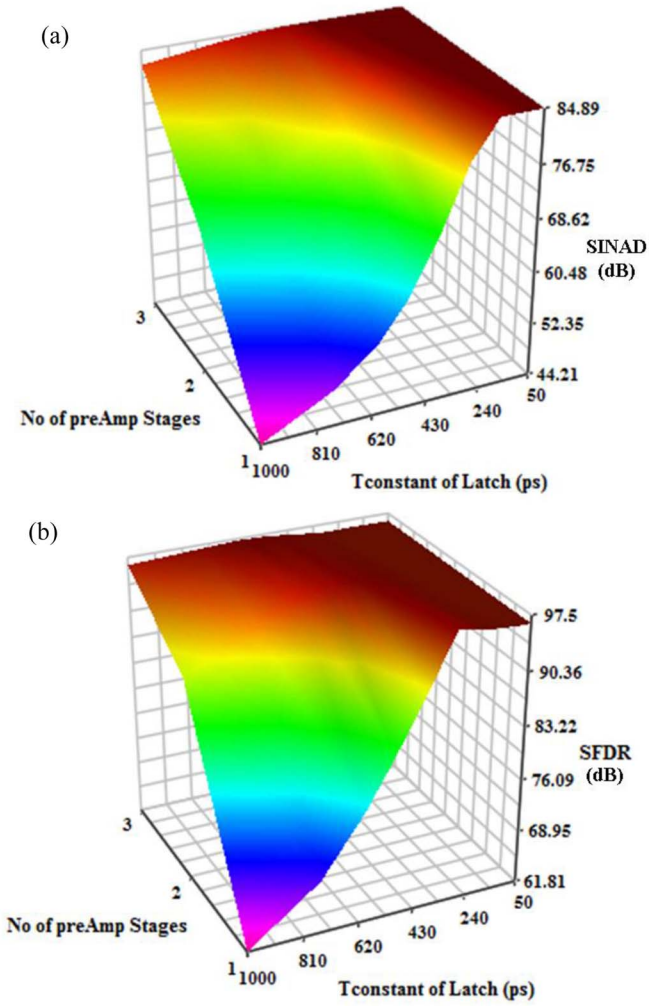


Fig. 14. (a) SINAD versus number of preamplifier stages and time constant of latch (τ_L). (b) SFDR versus number of preamplifier stages and time constant of latch (τ_L).

III. Analysis and Results

As a case study, a 14-bit, 4.2 MSPS SAR-ADC architecture is evaluated. Conventional static and dynamic tests have been done. In static testing, a dc analysis is done by applying a dc sweep voltage at input. In dynamic testing, a transient analysis is performed with sine-wave input. In postsimulations, standard parameters of ADC [35]–[37] are evaluated. For apple to apple comparison, all the simulations are done on SPICE netlist, structural model, and proposed modeling scheme (DPM).

A. Comparative Study of Sample and Hold

In general, performance of S/H circuit degrades with input-signal frequency (f_{in}) for settling issues. Comparative study of S/H circuit accuracy is done by looking at top-level performance with respect to f_{in}. For different f_{in}, signal to noise and distortion ratio (SINAD) and spurious-free dynamic range (SFDR) plots are shown in Fig. 12(a) and (b) (x-axis is normalized in terms of sampling frequency f_s). The pattern of SINAD and SFDR numbers came from proposed modeling scheme closely follows that of SPICE netlist.

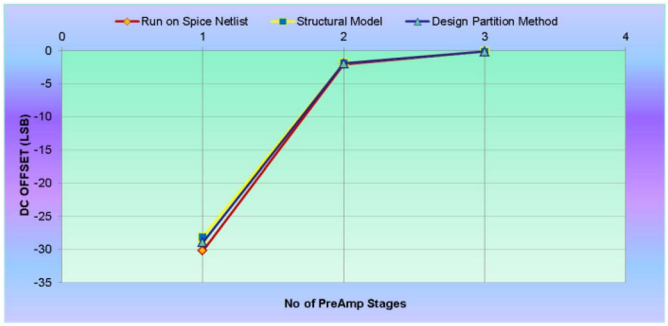


Fig. 15. ADC offset versus number of preamplifier stages for 100 mV of latch offset.

B. Comparative Study of Digital to Analog Converter

A hybrid DAC [20], [26] is used in the 14-bit SAR-ADC design (capacitive DAC for MSB bits and resistive DAC for LSB bits). Unit resistance (R_{unit}) and unit capacitance (C_{unit}) values for hybrid DAC are determined based on their mismatch numbers. σ_R and σ_C represent the SDs of the resistor mismatch and capacitor mismatch, respectively.

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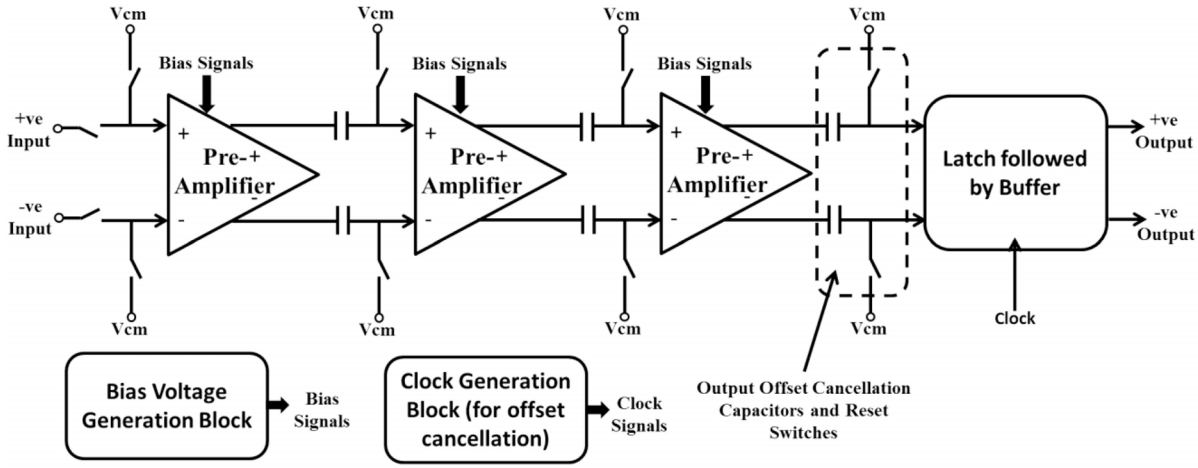


Fig. 16. Sub-blocks in the comparator schematic.

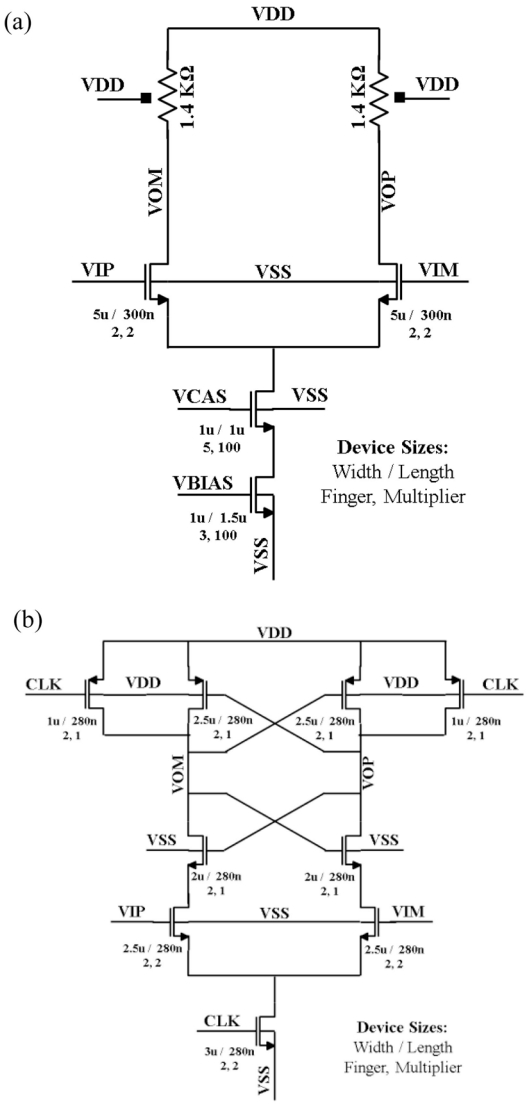


Fig. 17. (a) Schematic diagram of single-stage preamplifier. (b) Schematic diagram of the latch.

Worst-case SINAD and SFDR numbers from Monte-Carlo simulations for different σ_R and σ_C values are shown in Fig. 13(a) and (b). As expected, SINAD and SFDR degrade with higher σ_R and σ_C values. The specifications for the DAC are derived through the proposed modeling platform. R_{unit} value corresponding to 0.1061% σ_R and C_{unit} value corresponding to 0.0509% σ_C are chosen for final design. With those values, top-level performance is found to be satisfactory.

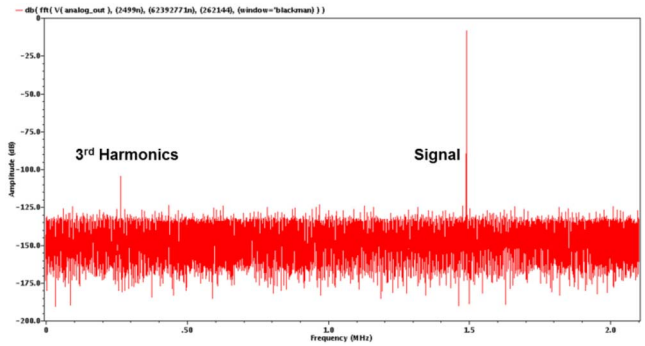


Fig. 18. Measured FFT plot from DPM.

C. Comparative Study of Comparator

In the proposed modeling scheme, top-level simulations are done with different number of preamplifier stages with different values of latch time constant (τ_L). For simplicity, voltage gains (Av) of all the preamplifier stages are kept same. The SINAD and SFDR results are shown in Fig. 14(a) and (b). The specifications of the comparator are also derived using the behavioral model of the comparator. Three-stage preamplifier along with a latch of $\tau_L \approx 80$ ps are used in the actual design. SPICE netlist simulation of the final top-level design reviles ADC performance is satisfactory with the derived comparator architecture.

Fig. 15 shows the plot of dc offset in SAR-ADC output for different number of preamplifier stages for 100 mV latch offset (intentionally applied). Here also results from proposed model closely follow the SPICE netlist simulation results.

Fig. 16 shows the sub-blocks of the comparator. Schematic of single-stage preamplifier and schematic of latch are shown in Fig. 17(a) and (b), respectively. Time constant of the latch is lesser than 80 ps in the worst corner.

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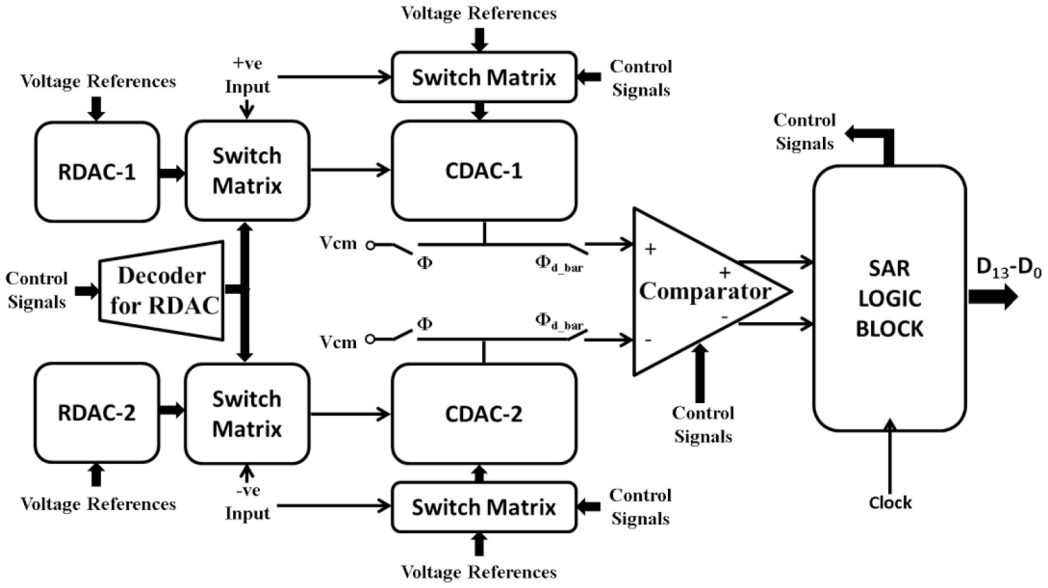


Fig. 19. Blocks in the SAR-ADC schematic.

D. Top-Level (SAR-ADC) Results

Fig. 18 shows a fast Fourier transform (FFT) plot of SAR-ADC output through proposed modeling scheme for 262 144 (2^{18}) FFT-points using "Blackman" window. SINAD (84.78dB) and SFDR (95.99dB) are achieved at top level of SAR-ADC.

Fig. 19 shows the blocks in the SAR-ADC schematic. Only major blocks are demarcated in the figure. The SAR logic generates the appropriate control signals based on bit by bit decisions.

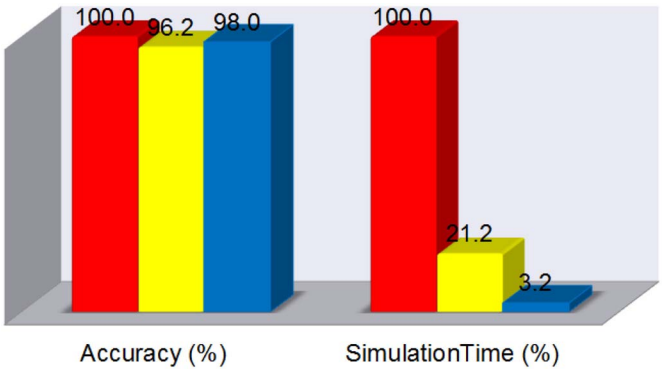


Fig. 20. Accuracy and simulation time (red: SPICE netlist, yellow: structural model, and blue: DPM).

IV. Computational Effort and Model Advantages

In the proposed modeling scheme, the whole SAR-ADC is realized through models of S/H, DAC, comparator, and SAR logic blocks. Here, S/H and DAC blocks are LUT based which results in fast top-level simulations. At the same time, data for LUTs of S/H and DAC are generated by simulating SPICE netlists with memory effects, which produces accurate data containing all nonidealities. At the beginning of the clock cycles, the block outputs are selected from LUTs. The modules at top level retain the selected outputs for the whole clock cycle. Dynamic response of comparator preamplifier is realized by a set of multivariable time-independent equations. The equations accurately calculate the final comparator output by solving algebraic equations only, which makes the model independent of iterations. SPICE netlists need iterations in continuous time as they work by solving differential equations. In the proposed model, iterations are required only at clock edges. This helps in reducing the computational effort.

Fig. 20 shows simulation time reduction in the proposed modeling scheme while providing better accuracy than the structural model. The accuracy numbers are evaluated by calculating percentage accuracy in terms of SINAD [Figs. 12(a) and 13(a)], SFDR [Figs. 12(b) and 13(b)], and offset measurement (Fig. 15), assuming SPICE netlist results are 100% accurate (reference results). Structural model lags in accuracy as individual blocks are not as accurate as SPICE-netlist while in the proposed modeling scheme; data for each individual block are generated by simulating SPICE netlist only. The simulation time of structural model is also seven times that of the proposed modeling scheme. The extremely fast top-level simulation (by eliminating the need of iteration) results in this simulation time reduction.

Time taken by the proposed modeling platform for a dynamic simulation (16 384 point FFT) is only ~3% of the time taken by equivalent SPICE netlist. Detail simulation time measurements are given in Table I. Effective simulation time is calculated by multiplying actual simulation time and number of simulator licenses.

The simulation time advantage becomes more pronounced in the cases where design changes are required in selective blocks only. Any changes in a single block results in regeneration of that block's data only. Table II describes the simulation time (13.5 h mentioned in Table I) breakup for different blocks. Similarly for large number of FFT points only S/H block and

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TABLE I
SIMULATION TIME COMPARISON

Analysis	Proposed Platform	Structural Model	Spice Netlist
Dynamic Testing (16384 FFT Points)	13.5 hrs	88 hrs	416 hrs

TABLE II
SIMULATION TIME BREAKUP

Analysis	Attributes	DAC	S/H	Top-Level
Dynamic Testing (16384 FFT Points)	Simulation Time	4.8 hrs	1.15 hrs	0.8 hrs
	Simulator Licenses	2	2	2
	Effective Time	9.6 hrs	2.3 hrs	1.6 hrs

TABLE III
MEMORY REQUIREMENT FOR LUT

Block Name	Data-Points	LUT Size (KB)	No. of LUTs	Memory Requirement (KB)
DAC	16384 (2^14)	~300	2	~600
S/H	16384 (2^14)	~300	2	~600
Total Memory Requirement (KB)				~1200

top-level need longer simulation times, same DAC data can be reused there. It also increases the reusability of the model.

Data in the LUTs for DAC as well as S/H are stored in μV order with six digits after decimal point, effectively the data accuracy is in the order of pV. As the LUT files contain arrays of numbers only, so the memory requirement is insignificant. As shown in Table III, for a 2^{14} point FFT simulation, ~1.2 MB memory is required for storing the LUTs.

In the case of Monte-Carlo simulation, mismatched element values are generated using random numbers. The values of SD for those random numbers are derived from the process design kit and Monte-Carlo simulation results. Then the mismatched element values are used to generate block-level LUTs. In multiple Monte-Carlo runs, different seed values are used to generate different sets of random numbers.

In the proposed modeling scheme, each functional block is realized separately in top level (DPM). In the case of undesired top-level performance, individual blocks can be made real while keeping other blocks ideal, which makes root cause analysis easy.

V. CONCLUSION

In DPM platform, the evaluation of different blocks across various levels of customization presents the perfect balance between accuracy, simulation time, root cause analysis, abstraction, cycle time reduction, and reusability. The platform is also very useful to perform sensitivity analysis of the different performance parameters. This aids in deriving the best possible optimization for block-level design specifications for a given technology process.

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